

ANKUSH PATIL

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Generative AI / Machine Learning Engineer with hands-on experience building LLM-powered RAG systems, transformer-based NLP models and FastAPI-based inference services. Skilled in PyTorch, Hugging Face, FAISS, Docker, embeddings, semantic retrieval and model evaluation. Familiar with LangChain, LangGraph and basic agentic workflows, along with AWS fundamentals for building scalable GenAI applications.

Technical Skills

Generative AI & LLMs: LLMs, Generative AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-tuning (LoRA/PEFT), Embeddings, Semantic Search, Model Evaluation, AI Agents, Agentic Workflows.
Frameworks & Libraries: LangChain, LangGraph, Hugging Face Transformers, PyTorch, SentenceTransformers, scikit-learn, NumPy, Pandas.
Vector Databases & Retrieval: FAISS, Chroma, Vector Search, Dense Retrieval, Document Chunking, Re-ranking, Similarity Search.
Backend & APIs: FastAPI, REST APIs, Python API Development, Inference Pipelines.
Cloud & Deployment: AWS (EC2, S3, IAM, basic deployment), Docker, GitHub Actions, MLflow.
Core ML / NLP / DL: Machine Learning, Deep Learning, NLP, Transformers, CNNs, Model Training, Evaluation.
Languages: Python, SQL, C++.

Academic Details

Year	Degree / Board	Institute	Marks/GPA
2021	B.E. Civil Engineering	K.K. Wagh I.E.E.R., Nashik	CGPA: 7.88
2017	HSC (Science)	K.T.H.M College, Nashik	73%

Work Experience

Site Engineer & Project Analytics Coordinator Feb 2023 – Present
Charwak Engineers & Developers, Nashik Onsite

- Analyzed government tenders, procurement records, BOQs and construction cost datasets to identify pricing trends, vendor patterns and budget optimization opportunities for project planning and execution.
- Maintained structured engineering datasets including soil investigation reports, material testing records, billing data, and project progress logs to support operational tracking, reporting and data-driven decision-making.

Projects

Agentic AI UPSC Preparation Platform [CODE] Feb 2026 – May 2026

- Tech:** LangChain, LangGraph, PostgreSQL (pgvector), FastAPI, Docker.
- Built an autonomous 6-node LangGraph pipeline to extract, draft, and format complex educational datasets, deploying specialized AI agents (Researcher, Drafter, QA Orchestrator) with zero human intervention.
- Implemented a dual-model LLM architecture, leveraging Qwen-72B for heavy cognitive tasks (factual research, 8-dimensional QA auditing) and Mistral-7B for fast, targeted tasks (distractor engineering, JSON formatting) to optimize cost and latency.
- Designed a RAG and self-correcting routing system using ChromaDB for semantic retrieval, automatically mapping QA failures back to specific nodes (up to 3 times) to guarantee unique data and zero hallucinations.
- Deployed the full-stack AI platform to production, serving the agentic backend via FastAPI REST endpoints to power a highly scalable, interactive web application featuring real-time analytics and gamified leaderboards.

Legal RAG System using LLMs (EUR-Lex) [CODE] Nov 2025 – Jan 2026

- Tech:** PyTorch, FAISS, SentenceTransformers, FastAPI, Docker, Hugging Face Hub
- Implemented FAISS-based dense retrieval with sentence embeddings over 57K+ EUR-Lex documents, reducing query latency and improving answer relevance in production RAG pipelines with FastAPI-based inference service.
- Designed comprehensive evaluation framework (Recall@K, Precision@K, MRR, latency analysis) and deployed Dockerized system with stateless API design, structured logging, and artifact hosting on Hugging Face Hub.

English→Hindi Neural Machine Translation using Transformer Architecture [CODE] Aug 2025 – Sep 2025

- Tech:** PyTorch, Custom Attention, Positional Encoding, Masking, Beam Search
- Built a complete Transformer-based neural machine translation system from scratch with multi-head attention, positional encoding, masking, and beam search decoding.
- Achieved BLEU score of 49.76, demonstrating strong sequence modeling and deep understanding of transformer internals.

Amazon Reviews NLP Pipeline [CODE] Jul 2025 – Aug 2025

- Tech:** Python, TF-IDF, Naive Bayes, BART, Hugging Face Transformers
- Developed large-scale NLP pipeline processing 3M+ reviews with sentiment analysis, aspect extraction, and BART-based summarization.

Leadership & Activities

Football Departmental Team Vice Captain: Led the departmental football team at K.K. Wagh I.E.E.R., Nashik during B.E. degree.